

Lab Sheet 06

Assumption 1: Linearity

The relationship between the independent and dependent variables should be linear.

Code

```
plot(Boston$lstat, Boston$medv, main = "Scatterplot of lstat vs medv",  
      xlab = "lstat", ylab = "medv")  
abline(lm(medv ~ lstat, data = Boston), col = "red")
```

Explanation

- A scatterplot shows the relationship between the predictor (lstat) and the response (medv).
- The red line is the fitted regression line. The points should follow a linear pattern.

Assumption 2: Independence

The residuals (errors) should be independent.

Code

```
library(lmtest)  
dwtest(lm(medv ~ lstat, data = Boston))
```

Explanation

- The Durbin-Watson test checks for autocorrelation in the residuals.
- A test statistic close to 2 indicates no autocorrelation.

Assumption 3: Homoscedasticity

The residuals should have constant variance.

Code

```
fit = lm(medv ~ lstat, data = Boston)
plot(fitted(fit), residuals(fit), main = "Residuals vs Fitted Values",
     xlab = "Fitted Values", ylab = "Residuals")
abline(h = 0, col = "red")
```

Explanation

- A residuals vs. fitted values plot is used to check homoscedasticity.
- The residuals should be randomly scattered around zero.
- Patterns (e.g., funnel shapes) indicate heteroscedasticity.

Assumption 4: Normality of Residuals

The residuals should be normally distributed.

Code

```
qqnorm(residuals(fit), main = "Normal Q-Q Plot")
qqline(residuals(fit), col = "red")
```

Explanation

- A Q-Q plot compares the distribution of residuals to a normal distribution.
- The points should fall along the red reference line.

Additional Test

```
shapiro.test(residuals(fit))
```

- The Shapiro-Wilk test checks for normality. A p-value > 0.05 indicates normality.

Assumption 5: No Multicollinearity

Independent variables should not be highly correlated.

Code

```
library(car)
```

```
vif(lm(medv ~ ., data = Boston))
```

Explanation

- Variance Inflation Factor (VIF) checks multicollinearity.
- VIF values > 5 indicate significant multicollinearity.